EXPERIMENT: 9

SIMULATION AND ANALYSIS OF UNIFORM AND NON-UNIFORM

QUANTIZATION USING MATLAB

AIM:

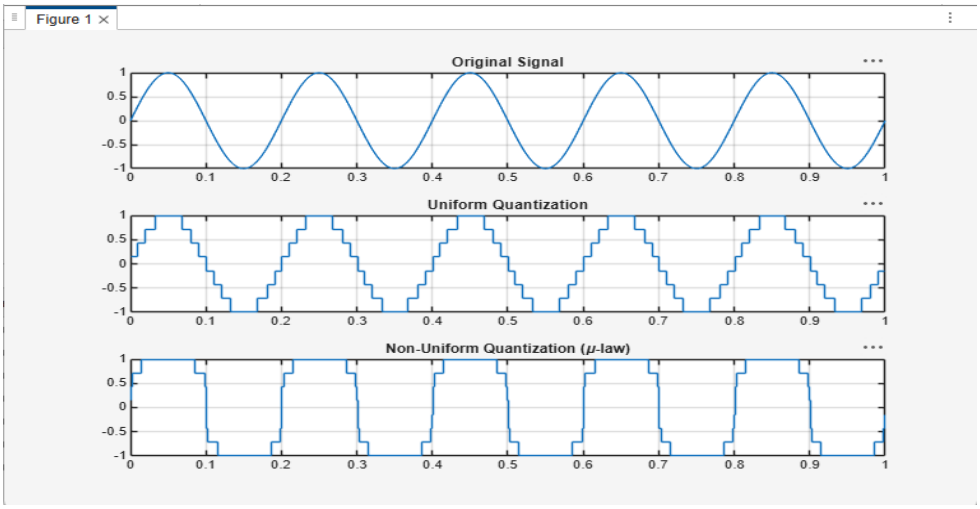
i) To implement and analyze uniform and non-uniform quantization techniques for an analog signal using MATLAB by converting the continuous-amplitude signal into discrete quantization levels.

ii) To compare the performance of uniform and non-uniform quantization by analyzing their quantized waveforms and quantization errors, and evaluate their effectiveness in representing signals with different amplitude variations.

CODE:

```
clc; clear; close all;
t=0:0.001:1;
x=sin(2*pi*5*t);
L=8; mu=255;
% Uniform Quantization
xu=round((x+1)*(L-1)/2)*2/(L-1)-1;
% Non-Uniform Quantization (μ-law)
xc=sign(x).*log(1+mu*abs(x))/log(1+mu);
xn=round((xc+1)*(L-1)/2)*2/(L-1)-1;
figure;
subplot(3,1,1); plot(t,x); title('Original Signal'); grid on;
subplot(3,1,2); stairs(t,xu); title('Uniform Quantization'); grid on;
subplot(3,1,3); stairs(t,xn); title('Non-Uniform Quantization (\mu-law)'); grid on;
```

OUTPUT:



RESULT:

i) The uniform and non-uniform quantization techniques were successfully implemented in MATLAB, and the corresponding quantized signals and quantization error waveforms were obtained.

ii) The performance comparison showed that non-uniform quantization produced lower quantization error for low-amplitude signals than uniform quantization, resulting in improved signal representation and better overall quantization efficiency.